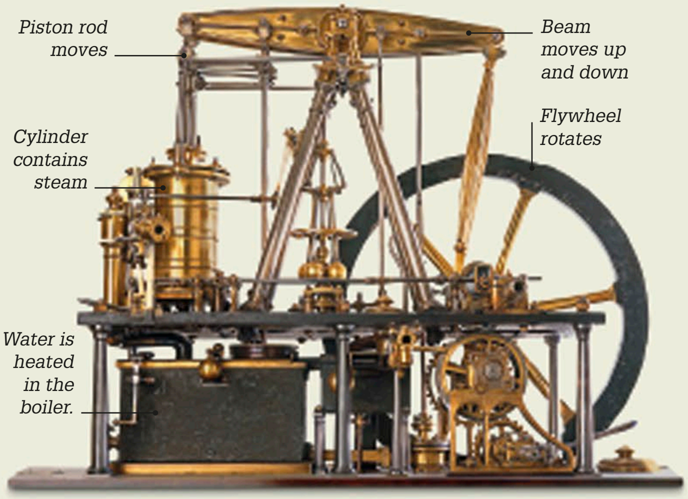
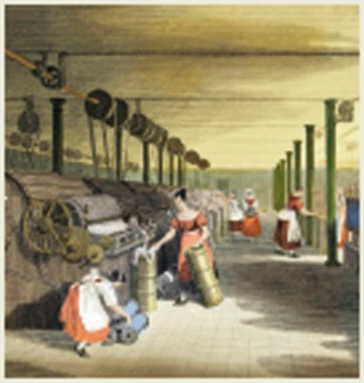
1760–1870 THE INDUSTRIAL REVOLUTION



The development of new technologies led to a period of social and economic change called the Industrial Revolution, when the growth of factories, mines, canals, and later, railroads, changed the landscape forever. Increasing numbers of men and women moved from the countryside to find work in the new industrial towns.

Getting up steam

In 1775, Scottish engineer James Watt designed a steam engine (see pp.130–131) that was smoother and more efficient than previous engines. His improvements meant that steam engines were no longer restricted to pumping water from mines, but could also be used to drive machinery in mills and factories, and to power steamships and railroad locomotives.



Textile mills

The invention of new technologies for spinning and weaving textiles led to the mass manufacture of cloth. Conditions were hard for the thousands of women and children who worked long hours in the new textile mills.

Women and girls at work in a cotton mill

1762 | 1766 | 1770

***Metrosideros collina*, one of the plants collected by Joseph Banks during his voyage**

1769

Scientific voyage

Captain James Cook voyaged to the Pacific in HMS *Endeavour* to observe the transit of Venus (when Venus passes in front of the Sun) from the island of Tahiti. He went on to discover the east coast of Australia in 1770. On board was botanist Joseph Banks, who collected thousands of plants in Botany Bay in Sydney, Australia. On their return, Banks's record of the voyage sparked interest across Europe.

1769

Steam car

French engineer Nicolas-Joseph Cugnot designed a steam-driven, self-propelled vehicle for carrying heavy weapons. It had three wheels and a large copper boiler that hung over the front wheel. The vehicle was so heavy that it proved impossible to steer.

In 1765, British physicist Henry Cavendish discovered that hydrogen, which he called "inflammable air," is a separate element.